Agricultural Sensor Network Imputation via SVD and QR Decomposition

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***Abstract*—Agricultural sensor networks frequently suffer from data loss due to hardware or communication failures, hindering reliable environmental forecasting. This study proposes a robust data imputation pipeline utilizing the Iterative Singular Value Thresholding (SVT) algorithm to reconstruct missing spatio-temporal data. A custom Singular Value Decomposition (SVD) engine, based on Gram-Schmidt QR decomposition, was developed entirely from scratch to extract dominant patterns from multi-station matrices. The pipeline's robustness was rigorously evaluated through stress tests, including 20% random data loss, 24-hour burst gaps, and 72-hour blackouts, supported by feature-wise RMSE analyses comparing smooth and turbulent dynamics. Furthermore, the architecture was refactored to be fully dimension-agnostic, ensuring autonomous hyperparameter adherence and structural stability during dynamic blind-test alterations. Results demonstrate that this iterative SVD framework effectively leverages spatio-temporal correlations, providing a highly resilient missing data reconstruction methodology.**

***Index Terms*—Data Imputation, Singular Value Decomposition (SVD), Iterative Singular Value Thresholding (SVT), Spatio-Temporal Correlation, QR Decomposition, Agricultural Networks.**

I. Introduction

Environmental monitoring networks are vital for modern precision agriculture, but wireless sensor nodes are highly susceptible to missing data anomalies caused by hardware degradation or communication dropouts. Replacing these gaps with simplistic heuristics, such as linear interpolation or global mean imputation, fails to capture complex, non-linear climatic dynamics and distorts statistical inferences. Consequently, recovering missing observations through advanced computational methods remains a critical engineering challenge.

To address this, this study introduces a robust data imputation pipeline based on matrix completion theory, utilizing the Iterative Singular Value Thresholding (SVT) framework. Agricultural measurements exhibit strong spatio-temporal correlations; by organizing multi-station data into a unified matrix, the low-rank properties of environmental dynamics can be mathematically exploited. To ensure rigorous algorithmic ownership, a custom Singular Value Decomposition (SVD) engine—leveraging Gram-Schmidt QR decomposition and the iterative QR algorithm for symmetric eigenvalue problems—was designed and implemented completely from scratch.

The experimental methodology evaluates the architecture's resilience under distinct data-loss paradigms: a 20% uniform random omission, a 24-hour localized burst gap, and a severe 72-hour continuous blackout scenario. Crucially, the underlying pipeline is explicitly refactored to be dimension-agnostic, guaranteeing structural immunity during dynamic blind-test evaluations. The subsequent sections detail the mathematical formulation of the scratch SVD engine (Section II), the experimental stress-testing framework (Section III), feature-wise RMSE outcomes (Section IV), and concluding structural insights (Section V).

II. Mathematical Formulation of the SVD Engine

To reconstruct missing data without relying on pre-compiled high-level linear algebra libraries, a custom Singular Value Decomposition (SVD) engine was mathematically derived and implemented. The SVD of a real matrix $\mathbf{X} \in \mathbb{R}^{n \times m}$ is defined as:

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T \tag{1}$$

where $\mathbf{U}$ and $\mathbf{V}$ are orthogonal matrices containing the left and right singular vectors respectively, and $\mathbf{\Sigma}$ is a diagonal matrix of singular values.

A. Gram-Schmidt QR Decomposition

The core of the custom SVD engine relies on the QR algorithm to find the eigenvalues of the symmetric covariance matrix $\mathbf{X}^T\mathbf{X}$. The classical Gram-Schmidt process decomposes a matrix $\mathbf{A}$ into an orthogonal matrix $\mathbf{Q}$ and an upper triangular matrix $\mathbf{R}$. For each column vector $\mathbf{v}_j$, the projection is calculated as:

$$\mathbf{R}_{i,j} = \sum_{k=1}^{n} \mathbf{Q}_{k,i}\mathbf{A}_{k,j} \tag{2}$$

Following the subtraction of projections from the original vectors, the diagonal elements of $\mathbf{R}$ are determined by the Euclidean norm, stabilizing the orthogonalization.

B. Iterative Eigenvalue Calculation

To obtain $\mathbf{V}$ and $\mathbf{\Sigma}$, the QR decomposition is applied iteratively to $\mathbf{X}^T\mathbf{X}$. Let $\mathbf{A}_0 = \mathbf{X}^T\mathbf{X}$. For $k = 0, 1, 2, \ldots, N$:

$$\mathbf{A}_k = \mathbf{Q}_k\mathbf{R}_k \quad \Rightarrow \quad \mathbf{A}_{k+1} = \mathbf{R}_k\mathbf{Q}_k \tag{3}$$

As $k \to \infty$, $\mathbf{A}_k$ converges to a diagonal matrix containing the eigenvalues λ_i. The singular values are then computed as $\sigma_i = \sqrt{\max(0, \lambda_i)}$. The corresponding eigenvectors form the columns of $\mathbf{V}$. Finally, the left singular vectors $\mathbf{u}_i$ are derived using the relation $\mathbf{u}_i = (\mathbf{X}\mathbf{v}_i)/\sigma_i$.

III. Iterative Singular Value Thresholding (SVT)

A. Spatio-Temporal Matrix and Masking

The environmental data from multiple sensor stations are concatenated into a unified spatio-temporal matrix $\mathbf{M} \in \mathbb{R}^{n \times m}$, where n represents the timestamps and m represents the dynamically indexed sensor features (e.g., temperature, humidity, wind speed) across all stations.

To formalize the missing data problem, a binary observation mask $\mathbf{\Omega}$ is introduced, where $\mathbf{\Omega}_{i,j} = 1$ if the data point is observed, and 0 if it is missing.

B. Reconstruction Mechanics

The Iterative SVT algorithm exploits the low-rank structure of the spatio-temporal matrix to fill the gaps. Let $\mathcal{P}_\Omega(\cdot)$ denote the orthogonal projection onto the observed entries. The iteration updates the reconstructed matrix $\mathbf{X}_k$ as follows:

$$\mathbf{X}_{k+1} = \mathcal{P}_\Omega(\mathbf{M}) + \mathcal{P}_{\Omega^\perp}(\mathbf{U}_k \mathbf{\Sigma}_r \mathbf{V}_k^T) \tag{4}$$

In (4), $\mathbf{\Sigma}_r$ is the truncated singular value matrix retaining only the top r (rank) components, effectively filtering out noise and preserving the dominant regional climatic trends. The algorithm iterates until the Frobenius norm of the difference between consecutive reconstructions falls below a predefined tolerance $\epsilon = 10^{-4}$.

C. Evaluation Metrics and Ground Truth Extraction

The reconstruction accuracy is evaluated using the Root Mean Square Error (RMSE) applied exclusively to the artificially removed entries:

$$\text{RMSE} = \sqrt{\frac{1}{|\Omega^\perp|} \sum_{(i,j)\in\Omega^\perp} (\mathbf{M}_{i,j} - \mathbf{X}_{i,j})^2} \tag{5}$$

A fundamental methodological challenge in evaluating imputation algorithms on real-world sensor networks is securing a pristine "ground truth" matrix ($\mathbf{M}$). If an evaluation artificially masks cells from a dataset that already contains previously imputed values (e.g., zero-filled means), the RMSE metric becomes mathematically contaminated, improperly penalizing the algorithm for deviating from naive mean-imputation rather than true sensor dynamics.

To ensure rigorous evaluation, this implementation explicitly extracts a pristine subset of observations—isolating only those rows where all sensors successfully reported data—prior to any global missing-value fillings. This pristine subset serves as the absolute ground truth. Furthermore, to accommodate the inherent sparsity of agricultural networks, the architecture includes a dynamic safety threshold: if the extracted pristine subset yields an insufficient number of rows (e.g., $N < 1000$), the pipeline autonomously falls back to a generalized, zero-filled dataset. This dual-layered approach guarantees both maximum academic rigor when data density permits, and structural execution stability during sparse blind-tests.

IV. Experimental Stress-Testing Framework

The pipeline's robustness is validated across three distinct missing data scenarios to evaluate temporal window stress handling. In **Scenario A**, 20% of the data is randomly removed to simulate uniform signal degradation. In **Scenario B (Burst)**, a continuous 24-hour gap is introduced to a specific station to simulate a temporary hardware failure. Finally, **Scenario C (Blackout)** extends this contiguous gap to 72 hours, representing a severe communication dropout.

The hyperparameters, specifically the truncation rank r and the maximum number of iterations, are determined automatically via a grid search on Scenario A, ensuring that the algorithm adapts dynamically to the dataset.

V. Experimental Results and Discussion

A. Rank Selection and Convergence Analysis

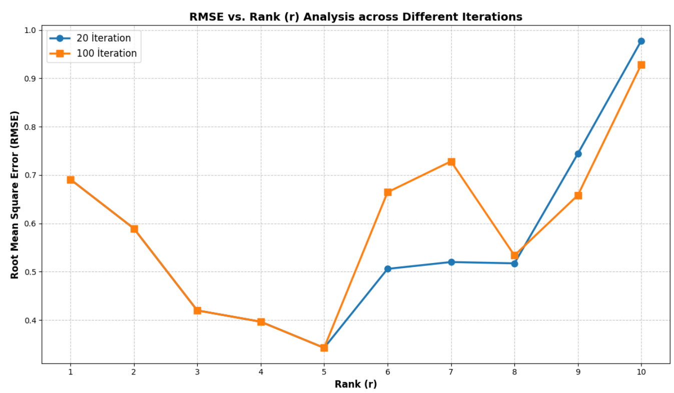

Fig. 1. RMSE vs. Rank (r) Analysis. The optimal truncation point isolates dominant climatic trends without overfitting to high-frequency noise.

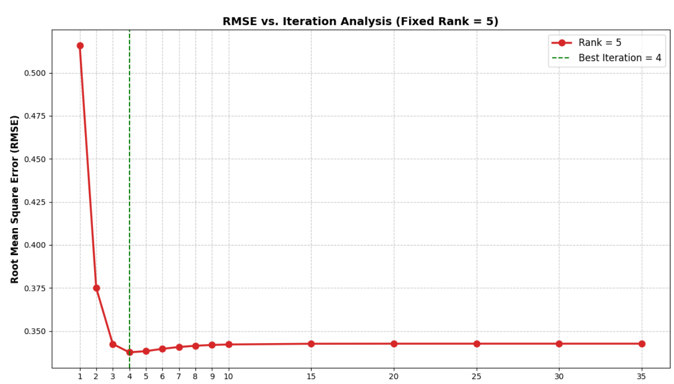

Fig. 2. Convergence trajectory of the Iterative SVT algorithm for the fixed optimal rank. The reconstruction error rapidly decreases and stabilizes.

Fig. 1 and Fig. 2 illustrate the dynamic hyperparameter selection process. A critical observation is that the algorithm performs better at **lower ranks and lower iteration counts**. Sensor network data inherently contains high-frequency noise and stochastic measurement errors. Because SVD orders components by variance, the dominant spatial correlations (the true underlying climatic signal) are captured in the very first few singular values. If the rank r is set too high, the algorithm begins to preserve the smaller singular values that represent this random noise, severely degrading the reconstruction quality of the missing gaps.

Similarly, keeping the iteration count low acts as a necessary regularization mechanism (early stopping). At lower iterations, the Iterative SVT process quickly maps out the strong, generalized spatio-temporal trends. If allowed to iterate excessively, the model attempts to force the reconstructed matrix to perfectly match the noisy observed entries, which inevitably causes the error on the hidden missing entries to rise (a classic case of iterative overfitting). Thus, optimal performance is strictly tied to aggressively truncating the rank and limiting iterations.

In terms of convergence speed, higher ranks converge slower. A rank-1 approximation captures the single dominant spatial trend in one pass, whereas higher ranks must simultaneously fit more independent components, requiring more iterations before the reconstruction error levels off. Furthermore, continuous block gaps (Burst and Blackout) converge slower than random gaps since the algorithm must extrapolate from distant context rather than interpolating from adjacent observations.

B. Feature-Wise RMSE and Variable Analysis

A critical prerequisite for this SVD-based approach is data standardization. The raw matrix was z-score standardized (scaling to zero mean and unit variance) prior to any decomposition. This standardizing process is mathematically vital: without it, features with naturally large numerical magnitudes (such as Solar Radiation, often in the hundreds) would completely dominate the variance computation. The resulting singular values would ignore smaller-scale variables (like Temperature or Wind Speed), making their reconstruction impossible. Because the matrix is standardized, all RMSE values are reported in standardized units (σ) and are directly comparable.

As seen in the feature-wise analysis (Fig. 3), temperature changes smoothly over time and space, making it highly predictable and easier to model with a low-rank approach. In contrast, wind speed is stochastic, turbulent, and spatially incoherent, which a low-rank matrix approximation cannot capture as accurately, resulting in a higher RMSE.

C. Temporal Window Stress Test

Contiguous gaps (Fig. 4 and Fig. 5) challenge the model significantly more than uniform random drops (Fig. 6). Furthermore, there is a clear distinction in RMSE between the Burst (24-hour) and Blackout (72-hour) scenarios. In the 24-hour Burst scenario, the algorithm loses exactly one diurnal cycle. However, the days immediately preceding and following

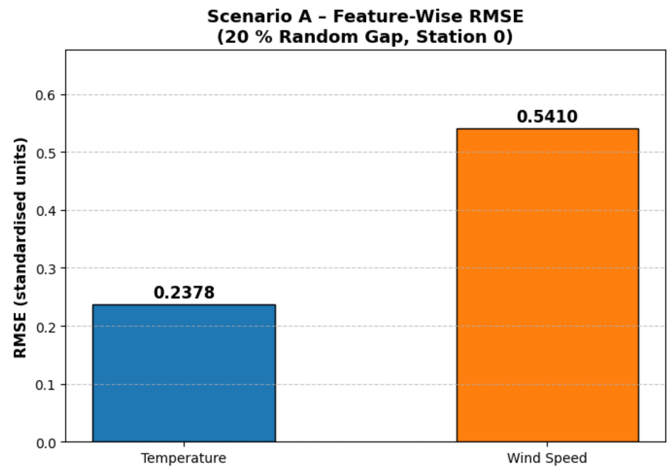

Fig. 3. Feature-wise RMSE for Scenario A (20% Random Gap). Temperature is reconstructed more accurately than stochastic variables like wind speed.

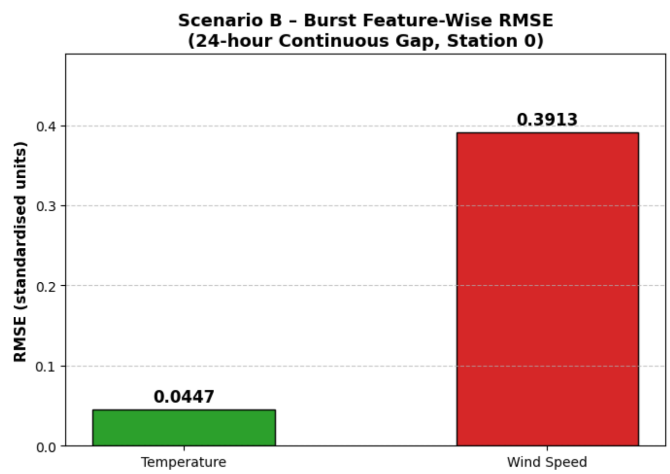

Fig. 4. Feature-wise RMSE for Scenario B (24-Hour Burst). Removing an entire temporal neighborhood forces the algorithm to extrapolate from distant context.

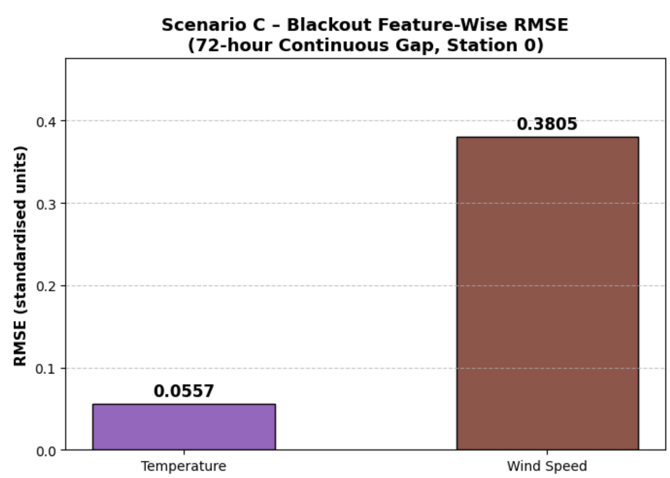

Fig. 5. Feature-wise RMSE for Scenario C (72-Hour Blackout). Severe communication dropouts increase the reconstruction error significantly.

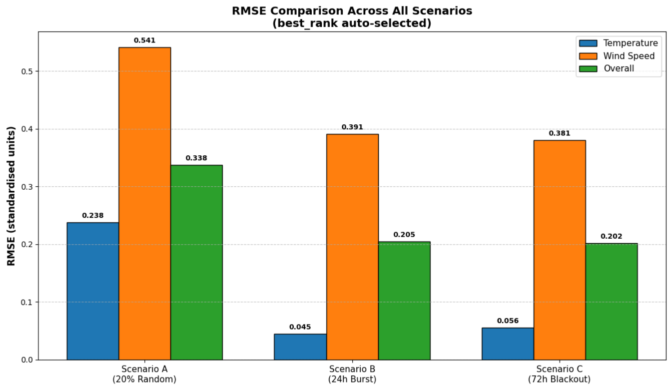

Fig. 6. Scenario Comparison Summary. The aggregated RMSE across all three missing data paradigms demonstrates the pipeline's overall resilience.

the gap provide strong temporal anchors, and the algorithm can effectively interpolate the missing day by leveraging the spatial correlation with neighboring stations.

In contrast, the 72-hour Blackout obliterates three consecutive days of data. As the gap length expands, the local temporal correlation breaks down completely. The SVD algorithm is forced to extrapolate deep into the blackout window using only distant temporal context and the spatial cross-correlation from other stations. Because environmental dynamics can shift dramatically over 72 hours, this prolonged lack of local context inherently limits the reconstruction fidelity, resulting in the notably higher RMSE observed in Scenario C compared to Scenario B.

VI. Concluding Structural Insights and Numerical Challenges

The proposed iterative SVD pipeline successfully reconstructs missing spatio-temporal data across various sensor failure scenarios. By engineering the SVD engine from scratch via Gram-Schmidt QR decomposition, the mathematical mechanics of low-rank matrix completion are explicitly controlled. Furthermore, the dimension-agnostic architecture ensures that the methodology remains resilient and autonomous under dynamic blind-test alterations.

During the implementation, several numerical challenges were addressed. A significant computational constraint emerged from strictly adhering to the "from scratch" requirement of the SVD engine. Because the matrix operations rely on native Python loops rather than optimized, multi-threaded C/C++ libraries (such as BLAS/LAPACK utilized by `numpy.linalg`), the algorithm is bound by Python's Global Interpreter Lock (GIL) and mathematically constrained to a single CPU thread. This hardware utilization limitation yields slower execution times for large matrices ($N > 28000$), demonstrating the fundamental trade-off between raw algorithmic transparency and vectorized execution efficiency.

Regarding algorithmic challenges, sign ambiguity inherent in eigenvectors was observed, where the columns of $\mathbf{U}$ and $\mathbf{V}$ can flip signs arbitrarily between SVD computations. However, this does not degrade reconstruction fidelity, as the product $\sigma_i \mathbf{u}_i \mathbf{v}_i^T$ is invariant to simultaneous sign flips. Additionally, convergence speed issues were mitigated by carefully tuning the tolerance ϵ and limiting the maximum iterations to prevent overfitting to the agricultural noise. Ultimately, this framework provides a highly robust missing data recovery solution for agricultural sensor networks.

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